

SERVEXA

Master Domain Discovery Blueprint

Enterprise Field Service Operations Platform

Purpose

Consolidated product charter plus the complete domain-discovery sequence for Servexa. This document freezes the working business model before PRD, UX, system architecture, data model and implementation.

Item	Value
Version	1.0 Candidate
Product	Servexa
Product class	Enterprise Field Service Management / Service Operations
Reference company	Prime Technical Services
Core stack direction	ASP.NET Core + React + SQL Server + AI/RAG
Architecture direction	Modular monolith first; explicit domain boundaries
Status	Domain discovery completed; implementation design not yet started

Prepared as a product/domain baseline. Competitor products are used only as capability benchmarks; Servexa retains its own vocabulary, workflows and architecture.

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0. Product Discovery Charter and Domain Principles

Servexa is a multi-tenant field-service operations platform for organizations whose technicians install, inspect, maintain and repair customer equipment at customer locations. The product is intentionally asset-centric and workflow-complete rather than a generic CRM or accounting ERP.

Product vision

Core lifecycle

Account → Site → Asset → Service Request → Work Order → Resource Requirement → Booking → Field Execution → Part Usage → Operational Completion → Billing → Payment → Asset History → Future Maintenance

Initial target industries

- Commercial solar installation and maintenance
- HVAC / chiller service
- Generator maintenance
- Elevator and facility maintenance
- Fire safety, CCTV and electrical service
- Industrial equipment and water-treatment service

Primary personas

Persona	Primary responsibility
Customer Contact	Request service, approve estimates, view appointments and invoices
Support Agent	Capture and triage service demand
Dispatcher	Match work demand to qualified resource capacity
Technician	Execute field work and capture evidence
Service Manager	Control service quality, scope and escalations
Warehouse / Procurement	Ensure parts availability and custody
Account Manager / Sales	Commercial relationship, estimates, agreements
Finance	Billing, payments, credits and financial closure
Branch / Operations Manager	Capacity, SLA, utilization and performance
System Administrator	Configuration, security, roles and integrations

Non-negotiable domain principles

- Business concepts must be modelled separately when their semantics differ; convenience must not collapse domain meaning.
- Physical work, inventory movement and financial charge are related but separate business facts.
- Work Order describes authorized work; Booking describes a scheduled visit; Resource Assignment describes who participates.
- Configuration and policies are preferred over tenant-specific hardcoding.
- Critical state changes are explicit, auditable and historically reconstructable.
- AI is advisory unless deterministic rules and explicit approval authorize action.
- Offline field execution is a first-class operational requirement, not a later UI enhancement.
- Modular monolith is the initial architecture direction; domain boundaries are preserved without premature microservices.

Workshop #1 — Customer, Site, Asset, Service Request & Work Order

Defines the identity of the customer, physical service location, installed equipment, incoming demand and the authorized operational unit of work.

Canonical concepts

Concept	Servexa meaning	Key distinction
Account	Legal/commercial entity with billing relationship	Not a Contact and not a Site
Contact	Human related to Account/Site	May be billing, onsite, approver, portal user
Site	Operational service location with access, hours, hazards and assets	Not merely an address
Equipment Model	Manufacturer/catalog template	Not a physical unit
Asset	Physical installed instance with identity/lifecycle	Not inventory SKU text on a job
Service Request	Unverified customer demand/symptom	May be resolved without field work
Work Order	Authorized scope of service work	May require multiple visits/resources
Operational Completion	Physical work finished	Financial work may remain
Financial Closure	Job locked for billing/costing	Does not mean physical work occurred now

Canonical lifecycle

Account → Site → Asset → Service Request → Triage → Work Order → one or more Bookings → Operational Completion → Financial Closure.

Key business rules

- D-101 Account, Contact and Site remain distinct.**
- D-102 Site is first-class and preserves history through ownership/billing changes.**
- D-103 Asset is distinct from Equipment Model and retains lifecycle/component history.**
- D-104 Service Request is demand; Work Order is authorized work.**
- D-105 A Work Order may cover multiple Assets at one Site when policy allows.**
- D-106 Work Order may require multiple visits and multiple resources.**
- D-107 Closed history is never silently rewritten; reopen/callback actions are explicit.**
- D-108 Operational Completion and Financial Closure are separate.**
- D-109 Cross-tenant entity relationships are prohibited.**
- D-110 Asset decommissioning is an authorized lifecycle action, not a technician free-form edit.**

Example

Shifa Hospital is the Account. Its Islamabad campus is a Site. Generator G-101 is an Asset. “Generator will not start” is a Service Request. “Diagnose and repair starting system” becomes a Work Order. Monday diagnosis and Wednesday repair are separate Bookings under the same Work Order.

Handoff to later workshops

- Booking/resource mechanics → Workshop #2
- Part/stock usage → Workshop #3
- Coverage/SLA/recurring service → Workshop #4
- Charges and invoice → Workshop #5
- Technician execution and offline behavior → Workshop #6

Workshop #2 — Scheduling, Dispatch & Workforce Management

Defines how work demand is translated into qualified, available capacity while preserving travel, territory, SLA, crews, overrides and real-time field status.

Canonical scheduling model

Scheduling chain

Work Order → Resource Requirement → Eligibility Filter → Candidate Scoring → Booking → Resource Assignment(s) → Dispatch → Travel → Arrival → Work

Core concepts

Concept	Meaning
Resource Requirement	Capability demand: roles, skills, certifications, quantity, duration, equipment
Resource	Schedulable person/crew/equipment; distinct from user account
Booking	Scheduled service visit/window for a Work Order
Resource Assignment	Participation of one resource in one booking
Skill	Operational capability, optionally with proficiency
Certification	Time-bound verified credential
Territory	Operational geography; distinct from organizational branch
Work Calendar	Shifts, breaks, leave, training, existing bookings, on-call/overtime
Scheduling Policy	Hard constraints + weighted preferences
Dispatch	Operational instruction to start/perform scheduled work

Frozen decisions

- D-201 One Work Order may have many Bookings.
- D-202 One Booking may have many Resource Assignments.
- D-203 Resource Requirement exists before selecting specific people.
- D-204 Eligibility and optimization/scoring are separate stages.
- D-205 Skills and Certifications are distinct.
- D-206 Branch and Territory are distinct.
- D-207 Availability is time/calendar based; travel consumes capacity.
- D-208 Pinned/confirmed bookings are not silently moved by optimization.
- D-209 Manual dispatcher authority is primary initially; optimization is recommendation-first.
- D-210 AI cannot determine deterministic availability, overlap or certification truth.

Eligibility vs scoring

Type	Examples	Effect
Hard constraint	Inactive resource, mandatory certification missing, tenant mismatch, impossible overlap	Candidate removed

Soft objective	Travel distance, preferred technician, skill depth, continuity, utilization	Candidate remains; score changes
Warning/override	Overtime, secondary territory, GPS uncertainty	Authorized user may continue with reason

State and exception model

Booking may move through Proposed → Scheduled → Confirmed → Dispatched → Traveling → Arrived → In Progress → Completed, with explicit Rescheduled, Cancelled, No Access, Paused and exception paths. Resource assignment history is preserved on substitution.

Example

A hospital chiller overhaul needs one lead HVAC technician, two additional technicians and a crane. Servexa creates one Booking for the customer visit and several Resource Assignments. Each person has individual travel and labour records while the customer still sees one appointment.

Workshop #3 — Inventory, Parts & Field Logistics

Defines product identity, physical custody, warehouse/truck stock, reservations, transfers, procurement, serialized items, consumption, returns and inventory auditability.

Critical separation

Never collapse these

Physical stock movement $\neq$ Work Order operational usage $\neq$ customer charge $\neq$ job cost.

Canonical concepts

Concept	Meaning
Product	Catalog identity for something purchased/sold/used
Inventory Item	Product whose physical quantity is tracked
Part	FSM usage context for an inventory item used on equipment
Inventory Location	Warehouse, van/truck, onsite store, transit/consignment location
Reservation	Physical stock still present but committed
Transfer	Custody movement between inventory locations
Consumption	Physical item used/removed in service execution
Serialized Unit	Individually tracked physical item
Lot/Batch	Group-tracked inventory identity
RMA/RTV	Controlled return and disposition process
Tool/Equipment	Reusable internal item; not consumable stock

Frozen decisions

D-301 Product, Inventory Item, Part Usage and Asset are distinct.

D-302 Inventory Location is first-class and may be mobile.

D-303 On Hand, Reserved, Available, On Order, In Transit and Quarantined are distinct states.

D-304 Reservation is not consumption.

D-305 Inventory truth is controlled by auditable movements/ledger entries, not unexplained quantity edits.

D-306 Serialized identity persists warehouse → truck → installation → return.

D-307 Warranty/discount never reverses real physical consumption.

D-308 Defective returns do not automatically become available stock.

D-309 Offline field consumption preserves physical reality and raises reconciliation exceptions when needed.

D-310 Ownership (company/customer/vendor consignment) is independent from physical location.

Example

Ali carries 5 control boards. One is reserved for a hospital job, leaving 4 generally available. He installs the reserved board: truck stock decreases, the Work Order records usage, the Asset gains a replacement component, Job Costing captures cost, and Billing independently decides whether the customer pays 0, discounted, or full price.

Workshop #4 — Contracts, Warranties, Entitlements, SLA & Preventive Maintenance

Defines the commercial service promise: who is covered, what benefits apply, timing commitments, warranty decisions and proactive recurring maintenance.

Canonical separation

Concept	Meaning
Service Agreement	Effective-dated commercial relationship with customer
Service Plan	Reusable/package-style service benefits adopted by an agreement
Entitlement	Contextual result of what this job is allowed/covered to receive
Warranty	Defect/workmanship coverage with validity and incident decision
SLA Policy	Timing obligations and escalation clocks
Maintenance Template	What technicians do during recurring service
Maintenance Plan	When/for which assets recurring maintenance is due
Maintenance Occurrence	One due instance generated by a plan

Frozen decisions

D-401 Agreement, Service Plan, Warranty, Entitlement, SLA and Maintenance Plan are distinct.

D-402 Agreement terms are versioned/effective-dated; historical jobs retain snapshots.

D-403 Entitlement is contextual and line-category aware (labour, parts, travel, fees, etc.).

D-404 Warranty active does not automatically mean this incident is covered.

D-405 Manufacturer claim/recovery is separate from customer service completion.

D-406 SLA uses separate milestones such as response, dispatch, arrival and resolution.

D-407 SLA uses business calendars, pause reasons, at-risk thresholds and escalation.

D-408 Maintenance Plan defines recurrence; Template defines work; Occurrence becomes Work Order.

D-409 Both fixed-calendar and completion-based recurrence must be supportable.

D-410 Agreement recurring billing is distinct from Work Order billing.

Example

A hospital Gold Plan covers labour and travel, gives 10% parts discount and a 2-hour arrival SLA. A generator failure is evaluated against agreement, asset coverage, warranty and current allowance limits. The resulting Entitlement Snapshot is attached to the Work Order so later contract amendments do not rewrite the historical decision.

Workshop #5 — Estimates, Pricebook, Billing, Job Costing & Payments

Transforms authorized/actual operational work into explainable prices, approvals, invoices, payments and profitability without turning Servexa into a full general ledger.

Domain objective

Servexa must answer two independent questions: “What should the customer pay?” and “What did this job actually cost us?” Revenue and cost converge in margin analytics but come from different sources.

Canonical financial chain

Commercial-to-cash

Pricebook → Estimate/Quote → Customer Approval → Work Order actual usage → Charge Calculation → Invoice → Payment/Credit/Refund → Financial Closure

Pricebook model

Concept	Meaning
Pricebook	Effective-dated pricing context by currency/segment/territory/customer
Price Item	Product/service price rule
Labour Rate	Hourly, role-based or skill-based service rate
Flat Rate	Fixed job/service fee independent of actual duration
Minimum Charge	Minimum billable amount/duration
Surcharge	After-hours, emergency, travel, hazardous work
Customer Price Agreement	Override/contract-specific pricing
Tax Rule Reference	Jurisdictional calculation input; tax engine/billing owns calculation

Mature FSM pricing supports effective-dated price lists, hourly rates, flat fees, duration rounding and minimum charges; Servexa adopts the business pattern while keeping its own rule model.

Estimate / Quote

Estimate is a versioned proposal of expected scope and price before final execution. It is not the Work Order and should survive rejection or revision without polluting operational history.

Estimate lifecycle	Meaning
Draft	Internal preparation
Pending Approval	Internal commercial approval needed
Presented	Sent to customer
Accepted	Customer authorized scope/price
Rejected	Customer declined
Expired	Validity ended
Superseded	Replaced by newer version

Estimate versioning

- Never overwrite an already-presented estimate silently.

- Each revision preserves previous scope, price, validity and approval evidence.
- Accepted version becomes the commercial baseline for authorized work where applicable.
- Additional work discovered in field creates a change estimate / scope-approval flow rather than editing accepted history.

Charge model

Calculation order

Actual operational usage → base price → contract/pricebook rules → entitlement coverage → approved discounts/surcharges → tax → invoice line.

The exact order is versioned/configurable but must be deterministic and explainable. Each invoice line retains source references to Work Order usage, estimate/pricebook and entitlement snapshot.

Billing bases

- Time and materials
- Flat-rate service
- Quoted/accepted fixed amount
- Per-unit service
- Recurring agreement billing
- Diagnostic/minimum charge
- Travel / mileage / zone fee
- After-hours or emergency surcharge
- Warranty-covered or zero-billed line with internal cost retained

Invoice domain

Concept	Key rule
Draft Invoice	Editable before posting/confirmation
Posted/Confirmed Invoice	Financially immutable except formal correction
Invoice Line	References commercial basis and source usage
Credit Note	Formal negative adjustment; does not erase original invoice
Refund	Payment-side return of money; separate from credit accounting
Void/Cancel	Allowed only according to lifecycle and integration state
Partial Invoice	Bills a subset/phase where policy allows

A completed Work Order can be operationally complete yet remain financially open until usage and coverage are reviewed. Once posted, silent mutation of billed usage is prohibited; corrections use credit/rebill or equivalent compensating records.

Payments

Concept	Meaning
Payment	Money received from customer
Payment Allocation	How one payment is applied across invoices
Deposit	Money received before final service/invoice
Unapplied Cash	Payment not yet matched
Refund	Money returned
Payment Status	Pending, Authorized, Captured, Failed, Refunded, etc. depending channel

Job costing

Job Costing captures the economic cost of fulfilling service, independent from customer price.

- Technician labour cost based on actual time/cost rate
- Inventory cost from consumed parts
- Travel/fuel/mileage cost where tracked
- Subcontractor cost
- Equipment rental / special tool cost
- Other reimbursable or internal expenses
- Warranty recovery expected/actual
- Allocated overhead optionally at analytics layer

Example

Customer invoice: Rs 90,000. Labour cost: Rs 12,000. Part cost: Rs 55,000. Travel cost: Rs 3,000. Gross contribution before overhead: Rs 20,000.

Agreement profitability

A preventive visit may have invoice value Rs 0 because the customer already pays a monthly/annual agreement fee. Job Costing still records the visit cost; analytics attribute agreement revenue and service cost to measure contract profitability.

Financial closure gates

- Required field execution evidence complete
- Actual labour/time reviewed
- Part usage reconciled
- Entitlement/warranty decision finalized enough for billing
- Customer-approved change scope reflected
- Charges calculated and exceptions approved
- Invoice posted or external accounting handoff accepted
- No unresolved financial-blocking exception

Key decisions

D-501 Pricebook is effective-dated and independent from Product master identity.

D-502 Estimate/Quote is separate from Work Order and versioned after presentation.

D-503 Actual work usage, pricing, entitlement and tax are separate calculation inputs.

D-504 Posted invoices are corrected through compensating documents, not silent edits.

D-505 Payment and payment allocation are separate from invoice creation.

D-506 Job cost is independent from customer charge.

D-507 Agreement profitability combines recurring revenue with service-delivery cost; zero job invoice does not mean zero cost.

D-508 Servexa remains service-financial subledger/operational billing; full general ledger can integrate externally.

D-509 Every calculated charge must be explainable back to source usage, price rule and entitlement.

D-510 Financial closure is explicit and later than operational completion.

Commands and events

Commands	Events
CreateEstimate / ReviseEstimate / PresentEstimate	EstimateCreated / EstimatePresented / EstimateRevised
AcceptEstimate / RejectEstimate	EstimateAccepted / EstimateRejected
CalculateCharges	ChargesCalculated / PricingExceptionDetected
CreateInvoice / PostInvoice	InvoiceCreated / InvoicePosted
IssueCreditNote	CreditNoteIssued
RecordPayment / AllocatePayment	PaymentRecorded / PaymentAllocated
RecordRefund	RefundRecorded
CloseWorkOrderFinancially	WorkOrderFinanciallyClosed

Important edge cases

- Estimate accepted but scope changes in field
- Customer provides part: labour billed, company inventory unaffected
- Warranty covers part but manufacturer later denies recovery
- Payment received before invoice
- One payment covers several invoices
- Invoice posted, then usage correction discovered
- Customer dispute without reversing physical service history
- Tax rules change after estimate but before invoice
- Multi-currency agreement or vendor cost
- Partial deposit and remaining balance

Workshop #6 — Field Execution, Inspections, Forms & Offline Mobile

Defines how technicians actually perform work onsite, capture trustworthy evidence and continue operating through unreliable connectivity.

Execution principle

The field app is not a smaller office UI. It is an operational instrument optimized for sequence, evidence, speed, safety and offline continuity.

Canonical technician journey

Field flow

Receive Assignment → Review pre-job brief → Start Travel → Arrive → Safety/access checks → Start Work → Diagnose → Tasks/Inspection → Labour/Parts/Readings/Photos → Additional Work Approval → Customer Acceptance → Complete Assignment/Booking → Sync

Execution session

Servexa should conceptually track actual participation per Resource Assignment: travel start/end, onsite arrival, work start/pause/resume/end, labour intervals and exceptions. Booking duration is the customer visit envelope; assignment execution is the individual worker truth.

Task vs Inspection vs Form

Concept	Meaning
Task	Action to perform, e.g. replace filter
Checklist	Simple set of required confirmations
Inspection	Structured data collection with typed questions, pass/fail and conditional logic
Form Template	Versioned reusable capture definition
Form Response	Immutable/controlled response instance for a specific job/asset
Reading	Measured value tied to asset/job/time and unit

Form capabilities

- Text, number, date/time, selection, yes/no, signature, photo, barcode/QR, measurement
- Required/optional questions
- Conditional branching
- Sectioning and repeatable groups where justified
- Validation ranges and units
- Pass/fail thresholds
- Evidence requirement on failure
- Template versioning; historical responses retain original schema
- Asset-specific or service-type-specific forms

Safety and gating

Some tasks/forms are informational; others gate progress. Example: a high-voltage isolation checklist may be mandatory before Start Work. Servexa must distinguish true invariant/compliance gates from configurable operational warnings.

Diagnosis and resolution

- Reported symptom remains separate from technician diagnosis.
- Root cause may be unknown, suspected or confirmed.
- Resolution/action taken is separate from diagnosis.
- Failure code taxonomy can support analytics but free-text evidence remains available.
- Repeat-failure/callback relationships should reference prior jobs/assets.

Photos, attachments and signatures

- Before/after photos can be required by work type.
- Each media item stores context: Work Order, Booking, Asset, Task/Inspection, author/time and optional location.
- Customer signature records acceptance/evidence, not necessarily legal waiver of every dispute.
- Signature artifact should be tamper-evident (document/version hash conceptually).
- Sensitive documents follow permissions and retention policy.

Offline-first domain behavior

Mature field-service mobile products cache necessary records and sync later. Servexa should explicitly model offline command identity, local timestamps, sync status, conflicts and server reconciliation instead of assuming always-online writes.

Offline concern	Servexa rule
Data scope	Download only authorized/needed future jobs, assets, forms and knowledge
Local commands	Every offline mutation receives unique command/idempotency identity
Timestamps	Preserve device-observed time plus server-received time
Conflict	Never silently overwrite contradictory server truth
Evidence	Photos/forms captured offline remain attached to original operational context
Security	Encrypted local storage/session controls; remote access revocation considered
Sync visibility	Technician sees pending/synced/failed state

Conflict classes

- Server cancelled job while technician started offline
- Dispatcher reassigned resource while old resource is offline
- Asset details changed while cached form references old version
- Inventory stock insufficient when offline consumption syncs
- Two offline edits update same non-mergeable field
- Form template updated after technician downloaded old version

Servexa preserves field evidence and raises explicit reconciliation where needed; it should not fabricate a clean history by discarding reality.

Field completion

- Individual Assignment completion can occur before Booking completion.
- Lead role may close visit after all mandatory resource/task conditions are met.
- Booking completion does not automatically financially close Work Order.
- Work Order may remain Paused / Awaiting Parts / Follow-up Required and generate another Booking.
- Customer refusal/additional-work decision is captured explicitly.

AI in field execution

AI may summarize prior service history, retrieve manual/SOP passages, draft technician notes, convert speech to structured suggestions and propose form values. Any update is permission-scoped and technician-confirmed; safety, warranty, inventory and billing truth remain deterministic.

Key decisions

- D-601 Field Execution is assignment/visit aware, not just Work Order status editing.**
- D-602 Task, Checklist, Inspection and Form Response remain distinct concepts.**
- D-603 Form templates are versioned; historical responses preserve their template version.**
- D-604 Symptom, Diagnosis, Root Cause and Resolution are separate operational facts.**
- D-605 Offline is a first-class state with idempotent commands and conflict handling.**
- D-606 Field evidence is preserved even when server state conflicts; reconciliation is explicit.**
- D-607 Customer signature/evidence is immutable/tamper-evident after final capture.**
- D-608 Assignment completion, Booking completion and Work Order operational completion may occur at different moments.**
- D-609 AI suggestions require human confirmation for operational updates.**
- D-610 Mobile permissions and downloaded data are minimal and assignment/context scoped.**

Core events

TravelStarted, TechnicianArrived, WorkStarted, WorkPaused, DiagnosisRecorded, ReadingRecorded, InspectionCompleted, SafetyFailureDetected, PartUsageRecorded, AdditionalWorkRequested, CustomerApprovalCaptured, CustomerRefusalRecorded, SignatureCaptured, AssignmentCompleted, BookingCompleted, OfflineConflictDetected, SyncCompleted.

Workshop #7 — Customer Portal, Communication & Notifications

Defines external self-service, customer visibility and the communication contract around service activity.

Portal principle

The portal is not an exposed internal back-office UI. It is a permission-limited customer experience built around the customer's own Accounts, Sites, Assets, Requests, Appointments, Estimates, Agreements and invoices.

Portal capabilities

- Authenticated customer access
- Account/Site-scoped user membership
- Create and track Service Requests
- View covered Assets and service history appropriate for external sharing
- View upcoming Bookings / arrival windows
- Self-schedule from safe slots where service type supports it
- Approve/reject estimates and additional-work requests
- View service reports and customer-visible documents
- View agreements/coverage summary
- View invoices, balances and payment state
- Manage notification preferences and contacts

Customer portal identity

Customer Contact and Portal User are related but not identical. A contact may have no login; a portal identity may represent a contact with explicit Account/Site access. Access can be revoked without deleting contact history.

Self-service scheduling boundaries

- Portal receives valid slot options from Scheduling; it never sees raw technician calendars.
- Complex crew/multi-resource jobs may require dispatcher review rather than instant booking.
- Customer preference is captured as a request/constraint, not direct internal resource manipulation.
- Temporary slot holds prevent two customers from claiming the same scarce capacity.
- Reschedule/cancellation rules come from service/contract policy.

Communication domain

Concept	Meaning
Communication Thread	Conversation context around Request/Work Order/Booking/Invoice
Message	Email/SMS/in-app/portal message record
Notification	System-generated event-driven communication
Template	Versioned reusable content with variables
Preference	Customer/user channel and opt-in policy
Delivery Attempt	Provider send attempt and status
Inbound Message	Customer reply converted into linked operational communication

Notification triggers

- Service request received
- Appointment proposed/confirmed/rescheduled/cancelled
- Technician dispatched / ETA update
- Technician delayed
- Estimate ready / approval reminder
- Additional work approval required
- Work completed / service report available
- Invoice issued / payment reminder
- Agreement expiring / renewal
- Preventive maintenance due
- SLA-sensitive customer update where policy requires

Communication safeguards

- Internal technician notes are private by default.
- Every externally rendered field/document must be explicitly customer-visible.
- Portal access is account/site scoped and tenant isolated.
- Notification retries must be idempotent to avoid duplicate SMS/email storms.
- Provider delivery status is not the same as customer reading/acknowledgement.
- Legal/marketing opt-in rules are configurable by region and channel.

Key decisions

D-701 Customer Contact and authenticated Portal Identity remain distinct.

D-702 Portal exposes curated customer-facing projections, not raw internal entities.

D-703 Self-scheduling consumes validated scheduling options; customers do not manipulate resources directly.

D-704 Internal notes/documents are private unless explicitly marked externally visible.

D-705 Notifications are event-driven records with delivery attempts, templates and preferences.

D-706 Communication thread preserves context across inbound/outbound channels.

D-707 Notification retry is idempotent and auditable.

D-708 Customer-visible service reports are generated from approved operational evidence.

D-709 Portal permissions can vary by Account, Site and role (requester, approver, finance, viewer).

D-710 Real-time technician location, if exposed, is policy-limited and only during relevant service windows.

Example

A facility manager opens a Service Request, receives a confirmation, selects one of three safe appointment windows, gets a technician-on-the-way notification, approves an extra Rs 18,000 repair from the portal, signs completion onsite and later downloads the service report and invoice. Internal margin, technician private notes and unrelated sites remain hidden.

Workshop #8 — AI, RAG & Enterprise Knowledge Architecture

Defines where AI adds legitimate operational value, what knowledge can be retrieved, how permissions and citations work, and which decisions remain deterministic.

AI principle

Authority rule

AI can retrieve, summarize, suggest and explain. It must not invent authoritative operational facts or silently execute high-impact business decisions.

Knowledge domains

Knowledge source	Examples
Technical knowledge	Manufacturer manuals, service bulletins, troubleshooting guides
Company knowledge	SOPs, safety procedures, playbooks, training documents
Customer knowledge	Site rules, contracts, warranties, special instructions
Operational history	Work Orders, diagnoses, readings, parts usage, inspections
Asset history	Failures, component replacements, meter trends
Commercial knowledge	Agreement clauses, price policies, entitlement explanations
Structured system truth	Schedules, stock, payments, SLA clocks - retrieved deterministically, not hallucinated

RAG pipeline

RAG flow

Document/record → parse → normalize → classify → chunk → metadata/security labels → embedding → vector/index store → permission-filtered retrieval → context assembly → LLM → cited answer → optional human-approved action

Security-before-retrieval

Tenant, Account/Site scope, user permission and document classification are applied before context reaches the model. “Retrieve everything then ask the LLM not to reveal it” is prohibited.

Knowledge metadata

- TenantId
- KnowledgeSourceType
- Document/Record identity
- Account/Site/Asset scope
- Security classification
- Effective/version dates
- Language
- Manufacturer/Equipment Model
- Document page/section/chunk locator
- Retention status
- Embedding model/version

Citation requirement

Answers based on documents should cite source document/page/section or system-record provenance. The user must be able to inspect the evidence behind troubleshooting, contract or procedural claims.

Core AI use cases

Use case	AI role	Human/deterministic control
Technician troubleshooting	Retrieve manuals + similar jobs; summarize likely checks	Technician diagnoses; safety rules remain deterministic
Pre-job brief	Summarize customer/site/asset/history/parts risks	Source records remain truth
Work Order summary	Condense progress, history, usage and next steps	Permission-scoped; summary is not a state change
Technician note assistant	Convert speech/rough notes into structured draft	Technician confirms before save
Inspection template drafting	Generate draft from manuals/SOPs	Admin reviews/publishes version
Natural-language analytics	Translate questions into governed metrics/query plan	Metrics engine executes deterministic query
Agreement explanation	Explain why coverage applied	Entitlement engine is authority
Inventory assistance	Suggest likely parts/replenishment	Inventory balances are authority
Scheduling explanation	Explain recommendation factors	Scheduling engine computes eligibility/score
Failure-pattern analysis	Synthesize repeat faults across history	Analyst validates causality

AI actions and tool use

Servexa should use an AI Gateway with explicit tools/functions. The model may request read operations freely within permission; write operations are classified by risk and usually require confirmation.

Action class	Examples	Policy
Read-only	Search manuals, summarize WO, query asset history	Allowed within permissions
Draft	Draft note, estimate narrative, customer message	Human review before publish
Low-risk write	Suggest/update non-critical note field	Explicit user confirmation
High-impact write	Change booking, entitlement, invoice, inventory adjustment	Deterministic validation + authorized confirmation; AI never bypasses rules

Model/provider abstraction

Domain code depends on interfaces such as LlmProvider, EmbeddingProvider and RetrievalService rather than one vendor. Initial local/open-weight provider can be replaced with hosted providers without redesigning business domains.

Vector/search architecture

SQL Server vector capability may be used initially behind an abstraction, but retrieval must support hybrid search (semantic + keyword/filter) and future replacement by a dedicated vector/search service if scale or quality requires it.

Prompt and model governance

- Version system prompts and tool schemas.
- Record model/provider/version for auditable AI outputs where material.

- Do not train/fine-tune on tenant data implicitly.
- Redact or exclude secrets and unsupported sensitive fields.
- Evaluate answer quality on representative field-service test sets.
- Support feedback: useful/incorrect/unsafe source.
- Apply response limits for safety-critical or uncertain answers.

RAG freshness

- Document version changes trigger re-indexing.
- Expired contracts/manuals remain historical but are not default-current evidence.
- Operational records can be indexed incrementally.
- Embeddings carry version so re-embedding migrations are controlled.

Key decisions

D-801 AI is an assistant layer; deterministic domain engines remain authoritative.

D-802 Permission filtering happens before retrieval/context assembly.

D-803 Document-based answers are source-cited and provenance-aware.

D-804 AI providers and vector stores are abstracted behind interfaces.

D-805 AI write actions are risk-classified and usually require explicit human confirmation.

D-806 Prompts, model versions and retrieval configuration are versioned for governance.

D-807 Structured truth (stock, schedule, SLA, payments) is queried deterministically, not guessed from embeddings.

D-808 RAG supports tenant/customer/asset/document metadata filtering.

D-809 AI-created inspection/checklist content remains Draft until authorized publication.

D-810 Evaluation and feedback datasets are part of AI product operations, not optional experimentation.

AI failure modes

- Hallucinated technical instruction
- Wrong/obsolete manual retrieved
- Cross-tenant retrieval leakage
- Overconfident warranty interpretation
- Invented stock availability
- Unsafe safety recommendation
- Stale asset history
- Duplicated AI action retry
- Prompt injection inside uploaded documents

Each failure mode requires guardrails at retrieval, tool validation, policy and UI explanation layers; safety is not delegated solely to prompt wording.

Workshop #9 — Reporting, Analytics, Automation & Integrations

Defines trustworthy operational metrics, configurable business automation and external-system boundaries that let Servexa operate as an enterprise platform.

Three distinct capabilities

Capability	Purpose
Reporting	Repeatable governed views of operational/financial facts
Analytics	Metrics, trends, segmentation and decision support
Automation	Event/condition-driven business actions with controls
Integration	Exchange commands/data/events with external systems

Metric governance

A KPI is a versioned business definition, not just a chart formula. Example: First-Time Fix Rate requires a precise definition of “first visit”, “resolved”, “callback window” and excluded work types.

Core KPI catalogue

Area	Representative KPIs
Demand	Service requests, backlog, priority mix, age
Scheduling	Utilization, unfilled requirements, travel ratio, overtime, reschedule rate
SLA	Response/arrival/resolution compliance, at-risk jobs, breach duration
Execution	First-time fix, repeat visit, job duration variance, checklist compliance
Assets	Failure rate, MTBF/MTTR, repeated components, maintenance compliance
Inventory	Stockouts, fill rate, turns, shrinkage, RMA rate, dead stock
Commercial	Estimate acceptance, average ticket, discount rate
Finance	Revenue, cost, gross margin, receivables, agreement profitability
Customer	CSAT/NPS if collected, cancellations, no-shows, complaints
Workforce	Productivity, utilization, skill coverage, certification expiry risk

Operational vs analytical models

Transactional domain models are optimized for correctness; dashboards often need read models or analytical projections. Servexa may build reporting projections/data marts without forcing domain aggregates to look like reports.

Automation engine

Automation structure

Trigger → Conditions → Action(s) → Approval/Policy → Execution → Result/Audit → Retry/Exception

Automation triggers

- Domain event (WorkOrderCompleted)
- Time-based schedule (daily/weekly)
- Relative deadline (SLA 80% consumed)
- Data change (Agreement expiring)
- Threshold (stock below reorder point)
- External webhook/event
- Manual user invocation

Automation actions

- Create task/request/Work Order
- Send notification
- Request approval
- Set priority/flag
- Create draft transfer/PO
- Generate report/export
- Call webhook/integration
- Create customer reminder
- Escalate SLA
- Queue AI draft/summarization

Automation safeguards

- Versioned rule definitions
- Dry-run/test mode
- Scope limits and rate controls
- Idempotency keys
- Approval gates for high-impact actions
- Loop detection
- Retry with dead-letter/exception queue
- Full run history: trigger, inputs, actions, outputs, failures
- Tenant isolation and permission-aware execution identity

Integration architecture

Servexa needs an Integration Hub/Adapter layer so domain modules do not embed vendor-specific logic. External systems are translated through anti-corruption mappings into Servexa concepts.

Integration type	Examples
Accounting/ERP	Invoices, payments, vendors, GL/export, product master depending authority
Payments	Payment gateway authorization/capture/refund
Maps/routing	Geocoding, ETA, routes
Messaging	SMS, email, WhatsApp/other approved provider
Identity	OIDC/SAML/SSO, enterprise directories
IoT/Telemetry	Meter readings, alerts, asset condition
Document storage	Object storage, e-signature provider
Tax/e-invoicing	Country-specific tax/e-invoice provider
External FSM/CRM migration	Customer/assets/work history import
Webhooks/API	Partner/customer system interoperability

System of record / authority mapping

Every integration must define who owns each fact. Example: Servexa may own Work Order and Asset service history while an accounting ERP owns posted general-ledger entries. Bidirectional sync without authority rules creates conflicts.

Integration reliability

- Outbox/inbox patterns for reliable event handoff
- Idempotent incoming webhooks
- Correlation IDs across requests
- Retry policy and dead-letter handling
- Versioned external mapping
- Replay capability where safe
- Partial failure visibility
- Manual reconciliation queue
- Rate-limit awareness
- Secrets rotation and least privilege

API principles

- Stable resource/action semantics aligned to ubiquitous language
- Tenant-scoped authentication and authorization
- Idempotency for retry-sensitive commands
- Pagination/filtering for lists
- Optimistic concurrency for mutable business resources
- Webhooks for important events
- Versioning/backward compatibility policy
- No direct database integration contract

Key decisions

D-901 KPIs have governed business definitions and versions.

D-902 Reporting projections are separate from transactional aggregate design.

D-903 Automation is explicit Trigger–Condition–Action with audit, idempotency and loop protection.

D-904 High-impact automation requires approval/policy gates.

D-905 External providers are isolated behind integration adapters/anti-corruption layers.

D-906 Every integration defines system-of-record authority per data domain.

D-907 Reliable outbound integration uses durable outbox-style handoff; inbound retries are idempotent.

D-908 Integration failures become visible reconciliation work, not silent data divergence.

D-909 Public/partner APIs follow domain language and enforce tenant authorization.

D-910 AI-generated analytics narratives never redefine governed KPI calculations.

Example automation

When AgreementExpiring occurs 60 days before end date, an automation verifies the agreement is active and auto-renewal is not already committed, creates an Account Manager renewal task, sends a configurable customer reminder, and records execution. If messaging fails, the task remains and the failed delivery is retried without duplicating the task.

10. Final Domain Consolidation and Cross-Domain Lifecycle

Servexa bounded-context map

Context	Primary ownership
Customer & Account	Accounts, Contacts, Sites and customer relationships
Asset Management	Equipment Models, Assets, components, asset lifecycle/history
Service Management	Service Requests, Work Orders, work scope/status
Scheduling & Workforce	Requirements, Resources, Booking, Assignment, dispatch/capacity
Field Execution	Tasks, inspections, readings, diagnosis, evidence, offline execution
Inventory & Procurement	Products, stock, locations, reservations, transfers, PO/RMA
Contracts & Entitlements	Agreements, plans, coverage, allowances
Warranty	Warranty records and recovery claims
SLA	Timing obligations, clocks, pauses, escalation
Preventive Maintenance	Plans, templates, occurrences
Commercial & Billing	Pricebooks, estimates, charges, invoices, credits
Payments	Payments, allocation, refunds
Job Costing	Actual fulfillment cost and margin
Customer Experience	Portal, communication, notifications
Knowledge & AI	Ingestion, retrieval, AI assistance/governance
Analytics & Automation	KPIs, projections, automations
Integration	External adapters, webhooks, sync/reconciliation
Identity & Platform	Tenant, users, roles, permissions, configuration, audit

End-to-end reactive repair lifecycle

1. Customer Contact submits a Service Request against Account/Site/Asset.
2. Support triages; entitlement and SLA candidates are evaluated.
3. Authorized Work Order is created with scope, requirements and snapshots.
4. Scheduling resolves eligible resources and commits one or more Bookings/Assignments.
5. Inventory reserves/picks/transfers likely required parts.
6. Technician receives offline-capable assignment package and pre-job knowledge.
7. Technician travels, arrives, diagnoses, performs tasks/inspections, records readings/evidence and part usage.
8. Additional scope, if any, creates estimate/approval flow; rejected work is recorded without corrupting original scope.
9. Booking/Work Order reaches operational completion or Paused/Follow-up path.
10. Asset history, component lineage and warranty/claim effects are updated.
11. Commercial engine prices actual usage using pricebook + entitlement + approved estimate/change scope.
12. Job Costing records labour/part/travel/subcontract cost.
13. Invoice posts; payment is recorded/allocated; exceptions use credit/refund flows.
14. Work Order reaches Financial Closure.
15. Reporting/analytics update; automation may trigger follow-up, renewal or quality review.

End-to-end preventive lifecycle

Maintenance Plan calculates due Occurrence from time/meter/condition rule.

Occurrence generates idempotent preventive Work Order within horizon.

Entitlement/Agreement identifies included services and SLA/service window.

Scheduling plans capacity; Inventory predicts/reserves standard kit.

Technician executes versioned checklist/inspection offline-capably.

Findings may generate corrective estimate/Work Order while PM itself completes.

Occurrence is completed; next fixed/rolling due date is calculated according to plan.

Cost is attributed to Agreement profitability even if job invoice is zero.

State authority summary

Question	Authoritative context
Who is customer / where is site?	Customer & Account
What physical equipment is installed?	Asset Management
What work is authorized?	Service Management
Who/when performs it?	Scheduling & Workforce
What actually happened onsite?	Field Execution
Where is the part physically?	Inventory
What is customer entitled to?	Contracts/Entitlement/Warranty
How fast was service promised?	SLA
What recurring work is due?	Preventive Maintenance
What should customer pay?	Commercial/Billing
What did work cost us?	Job Costing
Did customer pay?	Payments
What external message was delivered?	Customer Experience/Communication
What fact can AI cite?	Source domain + Knowledge provenance
What KPI is official?	Analytics governed metric definition

11. Decision Register, Open Questions and Discovery Exit Criteria

Cross-domain decisions that are now frozen as working baseline

- Account ≠ Contact ≠ Site.
- Equipment Model ≠ Asset ≠ Inventory Product.
- Service Request ≠ Work Order ≠ Booking ≠ Resource Assignment.
- One Work Order can have multiple Bookings; one Booking can have multiple Assignments.
- Physical stock, operational usage, customer charge and job cost are separate.
- Agreement, Service Plan, Entitlement, Warranty, SLA and Maintenance Plan are separate.
- Operational Completion occurs before/independently of Financial Closure.
- Estimate is versioned and separate from Work Order.
- Task, Inspection, Form Template and Form Response are separate.
- Customer portal is a curated external projection, not the back-office model.
- AI is advisory/tool-driven and permission filtered; deterministic domain engines remain authoritative.
- Reporting/automation/integrations are platform capabilities with explicit governance and audit.
- Modular monolith is the initial implementation architecture direction.

Open questions to validate before/while detailed design

Area	Open question	Current assumption
Crew	How often are crews static vs ad-hoc?	Support both; start with named crew + ad-hoc assignments
Vehicle	Is vehicle allocation mandatory for all tenants?	Optional schedulable/resource-linked capability
Asset ownership	Can owner differ from service/billing account?	Yes
Inventory valuation	FIFO/average/standard or external ERP?	Operational cost support; integration-friendly
Tax	Country-specific calculation provider/boundary?	Billing owns interface; jurisdiction adapter later
Agreement stacking	Can multiple agreements combine?	Policy-driven precedence and stacking
PM grouping	One WO per asset vs grouped site visit?	Configurable strategy
Offline merge	Which fields allow automatic merge?	Default conservative; explicit conflict for non-mergeable state
Portal self-schedule	Which job types are safe for instant booking?	Single/simple requirement first
AI deployment	Local open-weight vs hosted provider by tenant?	Provider abstraction supports both
Analytics warehouse	When to add dedicated OLAP store?	Start with read projections; evolve at scale

Domain Discovery exit criteria

Artifact	Status
Product Charter	Complete
Workshop #1 Core Service Domain	Complete
Workshop #2 Scheduling/Workforce	Complete at domain level
Workshop #3 Inventory/Logistics	Complete at domain level
Workshop #4 Contracts/SLA/PM	Complete at domain level
Workshop #5 Commercial/Billing/Payments	Complete at domain level

Workshop #6 Field Execution/Offline	Complete at domain level
Workshop #7 Portal/Communication	Complete at domain level
Workshop #8 AI/RAG/Knowledge	Complete at domain level
Workshop #9 Analytics/Automation/Integrations	Complete at domain level
Cross-domain lifecycle	Complete
Implementation architecture/data/API design	Intentionally not started in this document

Discovery verdict

PASS. Servexa now has a coherent enterprise domain baseline suitable for Product Requirements Definition, UX information architecture and architecture decision work. Domain validation with real field-service operators remains valuable and may refine policies without invalidating the core model.

12. Handoff Roadmap — What Comes After Domain Discovery

The next professional sequence should not jump straight to EF Core entities. Each stage consumes the frozen vocabulary and decisions from this document.

Stage	Primary output
1. Master PRD	Goals, personas, release scope, capabilities, acceptance criteria, non-functional requirements
2. UX Information Architecture	Navigation, role workspaces, screen inventory, user journeys, mobile/portal flows
3. Event Storming / Domain Validation	Validate commands, events, invariants and context boundaries with scenario walkthroughs
4. Architecture Decision Records	Modular-monolith module boundaries, tenancy, auth, offline sync, messaging, file storage, AI gateway, integrations
5. Logical Data Model	Entities/value objects, keys, effective dating, ledger models, concurrency boundaries
6. API & Integration Contracts	REST/action semantics, webhooks, idempotency, external adapters
7. Security Model	RBAC/ABAC, tenant/branch/site scope, approval model, audit classifications
8. Release Slicing	Foundation → Core FSM → Field app → Inventory → Commercial → Contracts → AI/analytics
9. Engineering Backlog	Epics, stories, acceptance tests, technical enablers
10. Implementation	Vertical slices with automated tests and observability

Recommended initial implementation slice

1. Tenant/Identity/permissions/audit foundation
2. Account + Contact + Site + Asset
3. Service Request + Work Order
4. Resource + basic calendar + Booking + Assignment
5. Technician field workflow online first with offline-ready command design
6. Basic Product/Truck stock and Part Usage
7. Pricebook + simple invoice calculation
8. End-to-end demo: Request → Dispatch → Service → Part → Signature → Invoice → Asset History

Only after this vertical slice is reliable should advanced optimizer, procurement, complex contracts, customer portal and AI be layered in.

Appendix A — Servexa Ubiquitous Language

Term	Definition
Account	Legal/commercial organization relationship.
Contact	Human associated with an Account or Site.
Site	Operational physical location where service occurs.
Address	Geographic/contact address value; not operational Site identity.
Equipment Model	Manufacturer/catalog definition.
Asset	Specific physical equipment instance with lifecycle and service history.
Component	Asset nested within/related to a parent Asset.
Service Request	Incoming unverified demand/symptom.
Work Order	Authorized scope of operational work.
Resource Requirement	Capabilities/quantity needed to perform a visit.
Booking	Scheduled customer/site execution window for a Work Order.
Resource Assignment	One resource's participation in one Booking.
Resource	Schedulable person/crew/equipment concept.
Technician	Human Resource performing field work.
Skill	Operational capability.
Certification	Verified/expiring credential.
Territory	Service coverage geography.
Inventory Location	Physical custody location for stock.
Reservation	Stock commitment without physical consumption.
Part Usage	Operational fact that an item was used in work.
Service Agreement	Versioned contractual service relationship.
Service Plan	Reusable/service-package terms adopted by Agreement.
Entitlement	Contextual coverage/benefit result for a service situation.
Warranty	Defect/workmanship coverage record.
SLA	Timing obligation and measured service target.
Maintenance Template	Versioned tasks/inspection content for recurring maintenance.
Maintenance Plan	Recurrence rule for preventive maintenance.
Maintenance Occurrence	One due instance generated by a plan.
Pricebook	Effective-dated pricing context.
Estimate	Versioned proposed scope and price.
Invoice	Customer billing document after charge calculation.
Credit Note	Formal negative correction to posted billing.
Payment Allocation	Application of received money to invoice balance.
Job Cost	Actual internal cost of fulfillment.
Operational Completion	Physical field work is finished.
Financial Closure	Work and billing/cost data are locked/reconciled sufficiently to close.
Task	Action technician must perform.
Inspection	Structured field data collection with rules/evidence.
Form Template	Versioned capture schema.
Form Response	Completed instance of a form.
Knowledge Chunk	Permission-labelled retrievable segment used by RAG.
Automation	Trigger-condition-action rule with audit/retry.
Integration Adapter	Boundary translating external system semantics into Servexa language.

Appendix B — Research Benchmarks and References

The following public documentation was used to validate that the major capability areas represented in Servexa are established Field Service Management patterns. Servexa's detailed domain model and decisions are its own working design, not a copy of any vendor implementation.

Reference	URL
Microsoft Dynamics 365 Field Service — price lists	https://learn.microsoft.com/en-us/dynamics365/field-service/create-price-list
Microsoft Dynamics 365 Field Service — posting work orders and invoice generation	https://learn.microsoft.com/en-us/dynamics365/field-service/post-work-order-invoice
Microsoft Dynamics 365 Field Service — customer agreements	https://learn.microsoft.com/en-us/dynamics365/field-service/setup-customer-agreements
Microsoft Dynamics 365 Field Service — inventory, purchasing and returns	https://learn.microsoft.com/en-us/dynamics365/field-service/inventory-purchasing-returns-overview
Microsoft Dynamics 365 Field Service — offline mobile	https://learn.microsoft.com/en-us/dynamics365/field-service/mobile/work-offline
Microsoft Dynamics 365 Field Service — offline synchronization	https://learn.microsoft.com/en-us/dynamics365/field-service/mobile/offline-data-sync
Microsoft Dynamics 365 Field Service — customer portal	https://learn.microsoft.com/en-us/dynamics365/field-service/customer-portal-overview
Microsoft Dynamics 365 Field Service — Copilot overview	https://learn.microsoft.com/en-us/dynamics365/field-service/copilot-overview
Salesforce Field Service — offline mode	https://developer.salesforce.com/docs/service/field-service-dev/guide/sfs-gs-lwc-fs-offline.html
ServiceTitan — Field Pro AI	https://help.servicetitan.com/docs/field-pro
ServiceTitan — Inventory software overview	https://www.servicetitan.com/features/contractor-inventory-software

Research interpretation

- Pricing/Field Service billing commonly derives invoice lines from used products/services and effective price lists.
- Offline mobile operation is a first-class requirement in mature FSM products; sync scope and conflict visibility matter.
- Customer portals support authenticated service requests, appointment visibility/self-service and technician-status communication.
- AI features in modern FSM systems increasingly include work-order summaries, natural-language assistance and inspection-template drafting.
- Inventory/vehicle stock, agreements, preventive maintenance, scheduling and portal capabilities are established market expectations; Servexa differentiates through coherent domain modeling, explainability, modular architecture and tenant-isolated AI/RAG.

Final Note

This Master Domain Discovery Blueprint intentionally stops before database schema, API contracts and React screen design. The next phase should preserve these business semantics and only then map them into technical structures. Any future change to a frozen decision should be recorded as a new domain decision with rationale and impact analysis.